# Proposition 8: fixes

Numbered checklist | 11 October 2026

Assessment: the stated limit is correct for a fixed sample; the proof needs the repairs below. The corrections are my mathematical review; the cited theorem supplies a general proof tool. No thesis files were changed.

Notation: $a = h_{min}$, $b = h_{max}$, $\delta = \delta_{ij}^{r,N}$, $\varepsilon = \varepsilon_N$. The empirical energy is written $Q_r$ below.

## 1. State the fixed-sample scope.

Fix $n \geq 1$ and the observations before taking $N \to \infty$. Keep r and the positive endpoints fixed. This proof does not establish a joint limit with $n = n(N)$.

Source: bandwidth-selection chapter, lines 768-775.

## 2. Correct the divergence domain.

Use $D(p,q) = \tfrac{1}{2}\|p-q\|_2^2$ for $p,q \in P \cap L^2$, or define D only on the Gaussian KDE family. The current P contains densities for which this divergence is infinite.

Source: bandwidth-selection chapter, lines 160-164; 769.

## 3. Repair the preceding grid-spacing lemma.

Write $h_j^r - h_i^r = r(j-i)\Delta_{r,N}$. Under $|j-i| = o(N)$, use $c_N = o(1)$ and remainder $O(\delta^2)$, not $O(N^{-2})$. Restore the missing 1/r in $Nc_N$. Explain o(N) as $|j-i|/N \to 0$. Taylor expansion of $g(u) = (ru)^{1/r}$ gives the desired result.

Source: bandwidth-selection chapter, lines 544-566.

## 4. Replace the integer-count calculation.

The bound is correct, but the displayed counting argument fails at integer endpoints. Use:

$$\#I_N(A) \leq 2\lfloor A/\varepsilon \rfloor + 1 \leq 2A/\varepsilon + 1.$$

Source: bandwidth-selection chapter, lines 781.

## 5. Correct the Fourier lower bound.

Use absolute values and integrate between min(h,s) and max(h,s). Replace the bandwidth/index typo. After squaring, the exponential is $\exp(-b^2\omega^2)$. A valid constant in $D(h,s) \geq c|u(s)-u(h)|^2$ is:

$$c = a^{4-2r}/(4\pi) \int_{\mathbb{R}} |P_n(\omega)|^2 \omega^4 \exp(-b^2\omega^2)\, d\omega > 0.$$

Source: bandwidth-selection chapter, lines 788-797.

## 6. Carry the tail normalization through every line.

The tail estimate concerns $\varepsilon T_N(A)$, not the unscaled $T_N(A)$. Fix the dropped or inconsistent ε factors. The final constant also loses a factor two:

$$\limsup_{N\to\infty} \varepsilon T_N(A) \leq 2\int_A^\infty \exp(-ct^2)\, dt \leq \exp(-cA^2)/(cA).$$

Source: bandwidth-selection chapter, lines 785-807.

## 7. Use absolute values in remainder estimates.

Divide by $|\tau_{N,j}|$ when it is nonzero. Bound the cross term in $|\rho_{N,j}|$ with $|\delta|$ times the two norms. For $j = i_N$, the remainders are zero; do not divide by zero.

Source: bandwidth-selection chapter, lines 815-827.

# Proposition 8: fixes

Checklist continued | items 8-14

### 8. Fill the first red gap by citing Taylor's theorem.

Set $u = h^r/r$ and let $F(u)$ be the Gaussian KDE at bandwidth $(ru)^{1/r}$. Check $F \in C^2$ in $L^2$ and $B = \sup\|F''(u)\|_2 < \infty$ on the fixed interval. A cited Taylor bound then gives:

$\|r_{N,j}\|_2 \le (B/2)|\delta|^2$; choose $a_N(A) = BA/(2\sqrt{\lambda_N}) \to 0$.

Source: bandwidth-selection chapter, lines 810-817. Taylor source: BYU Math 346, Theorem 6.6.9 and Corollary 6.6.14.

### 9. Fill the second red gap with a uniform estimate.

With $M = \sup\|F'(u)\|_2$, use the first remainder bound and $\lambda_N\delta^2 \le A^2$ on $I_N(A)$. This is the reason for the exponential error tending to zero; $\lambda_N/N^2 \to 0$ supplies the mesh condition instead.

$b_N(A) := A^2(Ma_N(A)+\tfrac{1}{2}a_N(A)^2) \to 0$;

$\sup_{j\in I_N(A)} \lambda_N|\rho_{N,j}| \le b_N(A)$, and $\sup_{j\in I_N(A)}|\eta_{N,j}| \le \exp(b_N(A))-1 \to 0$.

Source: bandwidth-selection chapter, lines 823-848.

### 10. Justify continuity and strict positivity of the energy.

Use the Fourier representation and Gaussian domination to establish continuity of the derivative in $L^2$. The integral below is positive because $P_n$ is continuous and $P_n(0) = 1$. This also verifies that the limit's denominator is nonzero.

$Q_r(h) = h^{4-2r}/(4\pi) \int_{\mathbb{R}}|P_n(\omega)|^2\omega^4 \exp(-h^2\omega^2)\, d\omega > 0$.

Source: bandwidth-selection chapter, lines 358-360; 850.

### 11. Explain why the local window avoids the endpoints.

Since $h_{i_N} \to h_0$ lies in the interior, both transformed endpoint distances multiplied by $\sqrt{\lambda_N}$ tend to infinity. Hence $I_N(A)$ is not truncated for fixed $A$ and large $N$. Also state $\varepsilon_N \to 0$ before the Riemann-sum step.

Source: bandwidth-selection chapter, lines 779; 850-862.

### 12. Replace the final equality by a squeeze argument.

For fixed $A$, the full row sum is not equal in the limit to the truncated integral. Put $L_A = \int_{-A}^{A} \exp(-t^2Q_r(h_0))\, dt$, use the bounds below, and only then let $A \to \infty$.

$L_A \le \liminf_N \varepsilon R_{i_N}^{(N)} \le \limsup_N \varepsilon R_{i_N}^{(N)} \le L_A + 2\int_A^{\infty} \exp(-ct^2)\, dt$.

Source: bandwidth-selection chapter, lines 864-870.

### 13. Make the remainder notation consistent.

Use $\delta_{ij}^{r,N}$ and $\tau_{N,j}$ throughout. Replace the inconsistent $\delta_N$, $\delta_{N,j}$, $\tau_N$, and $r_{N,j}$, and distinguish $i_N$ from an undefined $i$.

Source: bandwidth-selection chapter, lines 788-837.

### 14. Qualify the interpretation after the proof.

The red row-sum interpretation is asymptotic at interior bandwidths. Use $Q_r$, or explain $\beta = 2(1-r)$. "Slow change" refers to $u = h^r/r$; the first-derivative functional measures squared speed. The Perron-vector concentration in the subsection heading still requires a separate argument.

Source: bandwidth-selection chapter, lines 717; 874-883.

All line references: /Users/iljasukhanov/MASTERARBEIT/Abschlussarbeit_2021/LaTeX/chapters/5_bandwidth_selection.tex. Full derivations: output/analysis/proposition_8_review.md.